

Tania Morales

Statistician | [linkedin.com/in/tania-morales-s](https://www.linkedin.com/in/tania-morales-s) | Edinburgh UK tania.guieshuba@gmail.com |

PROFILE

Statistician and health data scientist with an MSc in Mathematics with Data Science and a background in radiotherapy treatment planning, genomics, and public health data analysis. Proficient in advanced statistical modelling, data management, and reproducible research, with experience curating and cleaning large-scale, complex datasets. Experienced in communicating analytical findings to academic, and non-specialist audiences, and in working across multidisciplinary teams in both academic and corporate settings. Currently tutoring postgraduate statistics and data science at the University of Edinburgh. Committed to developing expertise in linked NHS administrative data and clinical epidemiology, and eager to contribute to high-impact cardiovascular and renal health data research.

TECHNICAL SKILLS

Statistical programming: R (Tidyverse, RMarkdown), Python (NumPy, Scikit-learn, pandas, rapidfuzz, openpyxl), SQL, SAS, MATLAB, VBA, GitHub, LaTeX.

Statistical methods: Linear and logistic regression, tree-based models, Bayesian inference (JAGS), Monte Carlo simulation, time series forecasting, bootstrap methods, classification and machine learning. Committed to developing expertise in survival analysis and epidemiological study design.

Data management: Large scale data cleaning, standardisation and linkage. Controlled vocabulary development, structured documentation and developer handover

Visualisation & reporting: Plotly, Matplotlib, ggplot2. Statistical analysis plans. Regulatory and academic writing. Presentations to technical, and non-specialist stakeholders

Health data applications: Mortality and causes-of-death datasets, radiotherapy treatment optimisation, genomics and high-dimensional biological data, compositional data analysis.

RELEVANT EXPERIENCE

Data Analyst Volunteer Mar 2026 - present

Scottish Tech Army and Argyll & Sutherland Highlanders Museum

- Cleaned and standardised a large scale dataset of 24,000+ WWI soldier records using Python, resolving corrupted dates, a systematic lat/long coordinate swap affecting 5,437 records, and 1,001 regimental numbers corrupted by Excel serial conversion.
- Built a comprehensive data health check script covering 11 quality categories. Developed controlled vocabularies (100+ rank variants, causes of death, decorations). Implemented a fuzzy-matching duplicate detection system, and produced a structured developer handover pack with documentation for the development team.

Business Analyst Volunteer Feb 2026 - present

Scottish Tech Army and CrossReach, Edinburgh

- Completed a structured options analysis for a digital mental health platform, evaluating three delivery models with full cost justification (£5-50k), timeline defence to GDS standards, and a 40+ feature backlog, demonstrating evidence-based, independent analytical decision-making in a complex multi-stakeholder environment.
- Conducted market research using Edinburgh developer benchmarks and UK charity sector data (NCVO), and developed a detailed risk assessment covering GDPR, scope, and technology trade-offs to support stakeholder decision-making.

Data Science and Statistics Tutor Sep 2024 - present

The University of Edinburgh, UK

- Co-supervised a consultancy style MSc. project on fungal gene expression, advising on appropriate statistical methods for high-dimensional data.
- Reviewed model choice, assumptions and interpretation of statistical reports to ensure technical accuracy. Marked and gave feedback on over 400 assessments, focusing on reproducibility, modelling rigour, and interpretation. Mentored students in applying statistical thinking to real-world scenarios.
Notable courses: Statistical Programming with R, Machine Learning in Python, and Incomplete Data Analysis.

Data Scientist, Research Project (MSc. dissertation)

Feb 2023 – Oct 2023

University of Bath, England

- Implemented machine learning models on healthcare treatment data to optimise proton beam radiotherapy dose planning parameters, developing a treatment plan with minimised adverse effects using Python (pandas, Scikit-learn).
- Attended the MaThRad clinical workshop (London, 2023), engaging with clinicians and researchers on challenges in radiotherapy dose calculation. Presented findings in written report to an academic audience.

Applied Statistics Researcher (Undergraduate Thesis Research)

Jan 2021 – Nov 2021

Instituto Tecnológico Autónomo de México

- Analysed causes of death in Mexico using regression and compositional data analysis techniques applied to national mortality datasets, contributing to public health understanding. Produced a comprehensive academic thesis demonstrating independent research capability.

Senior Risk Manager

Jan 2016 - Jul 2018

GNP Insurance, Mexico City

- Built financial risk models (Monte Carlo simulation, forecasting) under Solvency II, improving the company's solvency margin by 20%. Produced risk reports for senior management, the board, and external regulators, translating complex technical analysis into clear recommendations.
- Managed a team of 5 analysts, assigning tasks, reviewing outputs, and responding to evolving business priorities, demonstrating the leadership capability.

Senior Financial Analyst

Aug 2013 - Dec 2015

GNP Insurance, Mexico City

- Oversaw modelling processes and aligned analytical outputs with regulatory requirements. Improved profitability forecasting and contributed to the company's Solvency II compliance programme.

EDUCATION

MSc. Mathematics with Data Science for Industry (Merit)

Sep 2022 - Oct 2023

University of Bath, England.

Modules: Statistical Modelling, Machine Learning, Bayesian Statistics, Mathematical Modelling.

BSc. (Hons) Actuarial Science (2:1)

Aug 2006 - Dec 2011

Instituto Tecnológico Autónomo de México, Mexico City.

Modules: Probability, Statistics, Stochastic Processes, Special Topics in Finance

Application for: **Epidemiologist / Data Scientist (UE07, Job ID 13762)**

Department: Institute for Neuroscience and Cardiovascular Research, University of Edinburgh.

SUPPORTING STATEMENT

Education and Qualifications

I hold an MSc in Mathematics with Data Science for Industry (Merit) from the University of Bath (2022-2023), with core modules in Statistical Modelling, Machine Learning, Bayesian Statistics, and Mathematical Modelling. My dissertation, in collaboration with the Mathematical Theory of Radiation Transport (MaThRad) network, applied machine learning and optimisation methods to proton beam radiotherapy treatment planning, a clinical research project that required translating a complex medical problem into quantitative analysis. My undergraduate degree is a BSc (Hons) in Actuarial Science (2:1) from the Instituto Tecnológico Autónomo de México, with strong foundations in probability, statistics, and stochastic processes.

Knowledge and Experience in Clinical and Epidemiological Research

My research background covers three distinct areas of health data science. For my MSc dissertation, I implemented machine learning models on theoretical data to optimise proton beam radiotherapy dose planning, developing a theoretical treatment plan with minimised adverse effects using Python (pandas, Scikit-learn). I attended the MaThRad clinical workshop in London in 2023, where I engaged directly with clinicians and researchers working on challenges in radiotherapy dose calculation. For my BSc thesis, I carried out independent research into causes of death in Mexico, applying regression and compositional data analysis techniques to national mortality datasets. I then presented a comprehensive academic thesis, demonstrating the capacity for independent, rigorous quantitative research.

More recently, as a tutor at the University of Edinburgh, I co-supervised a consultancy-style MSc project on fungal gene expression, advising students on appropriate statistical methods for complex, high-dimensional biological data. This experience is relevant to the genomics and biomedical data context of the Centre for Cardiovascular Science's research. I have also provided guidance on reproducibility and code quality to postgraduate students across courses in Statistical Programming with R, Machine Learning in Python, and Incomplete Data Analysis, marking and providing detailed feedback on over 400 assessments.

Data Management, Curation and Linkage

In a recent voluntary project with the Scottish Tech Army and the Argyll and Sutherland Highlanders Museum, I curated a dataset of over 24,000 records using Python (pandas, rapidfuzz, openpyxl). The project required me to identify and resolve a range of data quality issues: corrupted date fields across six columns, a lat/long coordinate swap affecting 5,437 records, 1,001 regimental numbers corrupted by Excel serial number conversion logic, surname corruptions, and impossible date values. I built a comprehensive data health check covering eleven categories of quality issues, developed controlled vocabularies across multiple fields (over 100 rank variants, causes of death, decorations, and religion),

implemented a fuzzy-matching duplicate detection system, and produced a developer handover pack.

While these are historical rather than clinical records, the methodological challenges are transferable to the linked NHS datasets. I am aware that working with patient data will require careful attention to information governance, and I am committed to developing this expertise.

Statistical Modelling and Programming

I am proficient in R (Tidyverse, RMarkdown) and Python, and have applied a wide range of statistical methods across research and professional contexts: linear and logistic regression, tree-based models, Bayesian inference (JAGS), Monte Carlo simulation, time series forecasting, bootstrap methods, and machine learning classification. In my senior risk management role at GNP Insurance, I built forecasting models under Solvency II using Monte Carlo simulation, achieving a 20% improvement in the company's solvency margin, demonstrating the application of rigorous statistical modelling to real-world, high-stakes decision-making. I have also worked with SQL, SAS, and MATLAB across various projects. I recognise that this role may involve survival analysis and specialised epidemiological methods beyond my current expertise, and I would be happy to develop these skills within a research environment.

Communication, Collaboration, and Mentorship

At GNP Insurance, I produced structured risk reports for boards of directors, senior management, and external regulators, translating complex model outputs into recommendations, and managed a team of five analysts, experience that supports the role's expectation to eventually help recruit and lead a team of health data scientists. In my volunteer business analyst role with Scottish Tech Army and CrossReach, I worked with stakeholders on a digital mental health platform, conducting market research, building a risk assessment, and presenting findings to a diverse stakeholder group. As a tutor at Edinburgh, I regularly adapt technical explanations to students, and I am experienced in working collaboratively within a research team while also progressing independently.

Motivation

This role brings together my skills, experience, and career goals. I am motivated by the chance to use advanced quantitative methods to help improve cardiovascular risk prediction and management for patients with kidney disease. Better data-driven insights could make a real difference for this group. I am confident I can make a valuable contribution to this program from the start, while also growing with the research group.

I live in Edinburgh and know the University's research environment well through my current tutoring role. My contract requires three months' notice, but with the semester ending soon, I could start earlier if needed. I have the right to work in the UK until December 2027. I would be happy to discuss my application further.

Best regards,

Tania Morales

April 2026